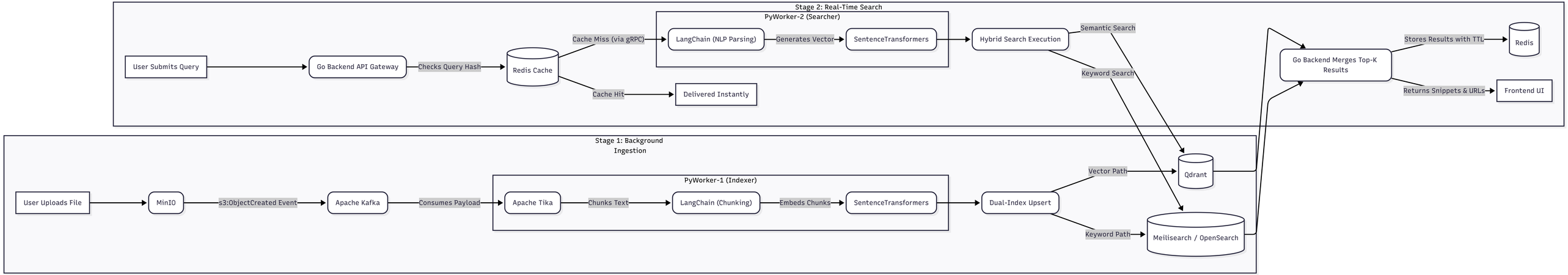
Stage 2: Real-Time Search
User Submits Query
Go Backend API Gateway
Checks Query Hash
Redis Cache
Cache Miss (via gRPC)
Cache Hit
Delivered Instantly
PyWorker-2 (Searcher)
LangChain (NLP Parsing)
Generates Vector
SentenceTransformers
Hybrid Search Execution
Semantic Search
Keyword Search
Go Backend Merges Top-K Results
Stores Results with TTL
Redis
Returns Snippets & URLs
Frontend UI
Stage 1: Background Ingestion
User Uploads File
MinIO
s3:ObjectCreated Event
Apache Kafka
Consumes Payload
PyWorker-1 (Indexer)
Apache Tika
Chunks Text
LangChain (Chunking)
Embeds Chunks
SentenceTransformers
Dual-Index Upsert
Vector Path
Qdrant
Keyword Path
Meilisearch / OpenSearch